Fieldnote

Source-linked answers for a changing oil field. A chat copilot joining inspection readings, repairs, report-only facts, operating rules and human context.

REHEARSAL DRAFT. Participant must review authorship/design and replace page 2 with the organizer's final-question results. This document contains sample evaluation results, not unseen-test claims.

Question → scoped plan → effective records + source rules → deterministic tools / bounded SQL → inline visual + evidence

Storage, indexing and retrieval

Node.js 24 and SQLite/FTS5. Originals are retained by record ID and path; per-well indexes join inspections, repairs and regulator events. Report files are followed through report_file rather than guessed filenames. Manuals, SOPs and repair reports are indexed. Additional datasets are validated and loaded alongside the sample; identical records deduplicate and conflicting IDs fail preflight.

Reasoning and tools

Gemma produces schema-constrained plans. Literal well IDs, dates, scopes and limits are checked in code, with adjustments recorded. Domain tools calculate production from manual temperature deltas, downtime from weekly inspections, corrected spill statistics, repair costs, RCA and compliance. Report containment and interim status come from text. Statistical questions can use a read-only SQLite worker over effective public facts, with provenance IDs, a row cap and timeout. Document answers cite retrieved source clauses.

Memory and conflicting evidence

Append-only shared corrections preserve original/previous effective values, actor, reason, date and inspection-field scope. Policies match either one well or a whole type plus equipment. Private conversation history is partitioned by user and persists across chats. Ordinary claims never overwrite measurements. Competing statements remain visible; unsupported current-state claims are marked unconfirmed. Report assertions remain historical when corrections invalidate their basis.

Generative interface and resilience

The answer envelope renders a status card, table, line chart, bars, geographic map or heatmap inline. Explicit visual requests win. Every answer exposes sources, dates, calculation plan and model usage. Supported local operations and exact SOP clauses remain available during model outages, with an explicit fallback label. The challenge user picker is not production authentication.

Model, cost and next steps

Default model: gemma-4-26b-a4b-it on Google's API; OpenAI-compatible endpoints are configurable. This recorded run contains 22,544 reported input and 728 output tokens from 27 successful calls. Failed requests may not return usage. Gemma is described as free-tier by the supplied guide; invoice cost is not measured. Next: wider paraphrase/property tests, stronger semantic SQL validation, authenticated deployment and incremental imports.

Application: MIT. Dataset/manuals/SOPs: ©2026 FlytBase, MIT, github.com/FlytBaseAILabs/flytbase-ahc-swe-asset-intelligence-dataset. AI-assisted development using OpenCode; the participant is responsible for understanding and presenting the work.

Answers you can inspect.

Analysis date: **2026-10-17**. Run: 2026-10-17T08:33:30.238Z. Transport: HTTP /api/chat. Each scenario starts from original sample data and fresh memory, then executes its specified prerequisite turns.

18/18
scenario assertions passed

27
model-backed turns

4
explicit local fallbacks

Case	Turns	Result	Observed final answer excerpt
Q1	1	**PASS**	WELL-0001: No, not producing oil. Jake Morales: Well shut in from 7 Oct for planned rod string replacement. Work order n…
Q2	1	**PASS**	4 wells not producing oil.
Q3	1	**PASS**	13 readings. Not producing on WELL-0001: 8 Oct 2026; WELL-0001: 15 Oct 2026.
Q4	1	**PASS**	6 real spills, 15.1 barrels total. 1 corrected false detections excluded.
Q5	1	**PASS**	WELL-0001: 21 days. Each nonproducing inspection counts as 7 days; missed inspections add zero.
Q6	1	**PASS**	WELL-0121: 49 days · WELL-0107: 42 days · WELL-0138: 42 days. Each nonproducing inspection counts as 7 days; missed insp…
Q7	1	**PASS**	4 real spills, 10.2 barrels total. 2 corrected false detections excluded.
Q8	1	**PASS**	141 real spills by geographic area and month. Corrected false detections removed.
Q9	1	**PASS**	4 breaches; USD 5,000 recorded fines, USD 5,000 still open. Pending actions are listed separately from breaches.
Q10	1	**PASS**	2 real spills, 1.1 barrels total. SOP SR-02 containment limit: 4 hours. Times come from repair reports.
Q11	2	**PASS**	WELL-0107: USD 5,176 · WELL-0121: USD 25,794 · WELL-0138: USD 46,942. Costs selected by repair start date; open-job cost…
Q12	2	**PASS**	3 wells not producing oil. Excluded after correction: WELL-0164 is producing (21.1 °C above ambient). wellhead_temp_c co…
Q13	1	**PASS**	Correction saved against INS-0151-0201: original 4.5, corrected 5.4, by Michael Turner. reportable spill. No written not…
Q14	3	**PASS**	4 wells/equipment groups require root-cause analysis. Michael Turner's rule for T350 pump_jack; team policy, not the man…
Q15	4	**PASS**	1 real spills, 5.4 barrels total. 2 corrected false detections excluded.
Q16	5	**PASS**	I have no such earlier conversation with Jessica Reed. Which wells do you mean? Private conversations belonging to anoth…
Q17	3	**PASS**	Saved shared note by Jake Morales as unconfirmed. The 2026-10-15 inspection by Ryan Brooks shows 3.2 °C above ambient: n…
Q18	1	**PASS**	Yes, the oil detection stands: 1 bbl (INS-0101-0197). Ordinary feedback does not override a detection; only a formal cor…

What worked, what remains

Source joins, corrections, shared/private memory, conflict handling and the prescribed visual shapes are checked against the supplied scenarios. Literal-constraint validation is part of these results; this is not a test of an unguarded model. Provider failures occurred during development and motivated visible local fallback. Broader numerical reasoning remains model-dependent; not every operating obligation is provable from incomplete records.

Final-test status: the final dataset and questions have not been supplied in this workspace. Import them, run the actual questions, export the responses and replace this rehearsal page before submission. Record the required own-voice video and publish the licensed repository.

Evidence/transcripts: results/live-evaluation.json. Exact expected answers live only in tests, never in runtime retrieval. UI screenshots are separately labeled local-mode captures.